CHISHOLM CATHOLIC COLLEGE

Year 11 Mathematics Methods

Test 1 2025

Total Mark
/49

Section 1: Calculator - free

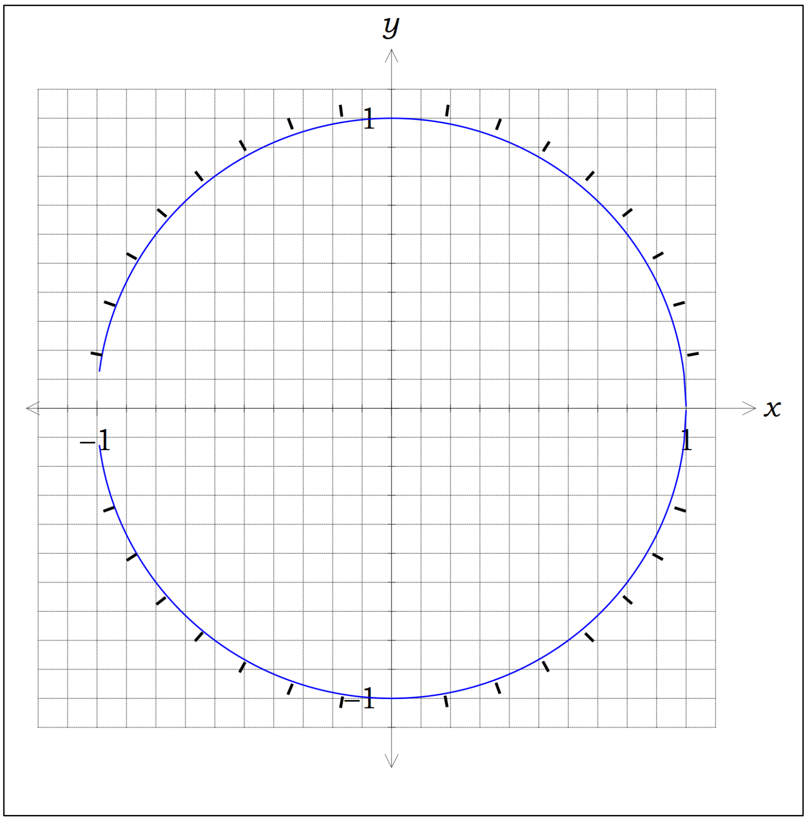
Name:	Time: 25 minutes maximum
Teacher:	Marks available: 22

Instructions:

- Answer all questions in the spaces provided.
- Calculators and notes are not permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.

Question 1 **[2, 2 = 4 marks]**

a) Use the unit circle below to determine the value of (to 2 decimal places if necessary):



(i) $\sin 70°$

(ii) $\cos \frac{3\pi}{2}$

b) Determine the size of the angle(s):

(i) $\tan(\theta) = 1$

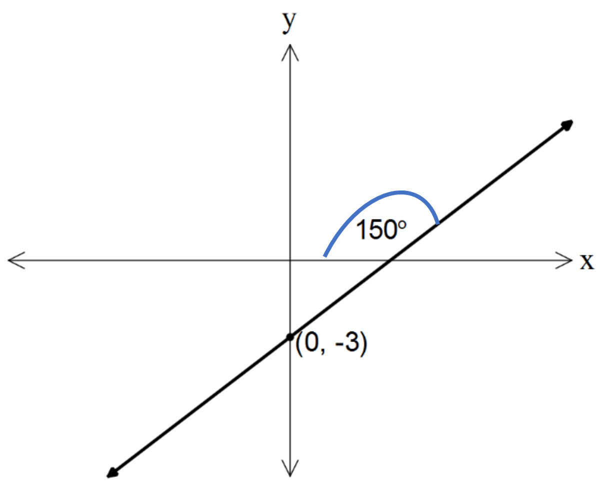
(ii) $\cos(\theta) = \frac{-1}{2}$

Question 2 **[1, 1, 2 = 4 marks]**

a) Write 225° as an angle in radians in simplest form

b) Convert $\frac{7\pi}{3}$ to degrees.

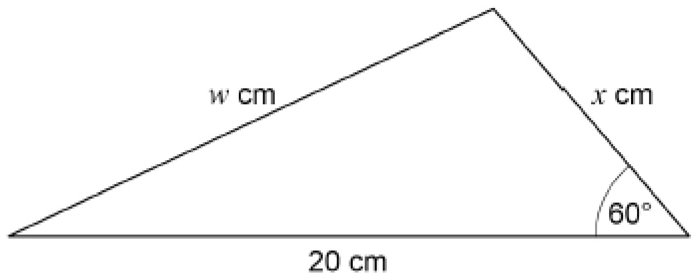
c) Write down the angle of inclination of the line below and hence determine the gradient of the line as an exact value.



Question 3 **[2, 3, 2 = 7 marks]**

The area of this triangle is $25\sqrt{3}$

Diagram not to scale



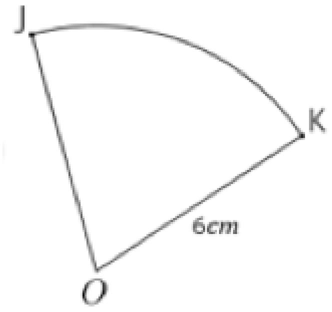
a) Show that the length of x is 5 cm

b) Find the value of w. Give your answer in the form $a\sqrt{b}$ where *a* and *b* are integers greater than 1

c) Show that the value of cos 30° × tan 60° + sin 30° is an integer.

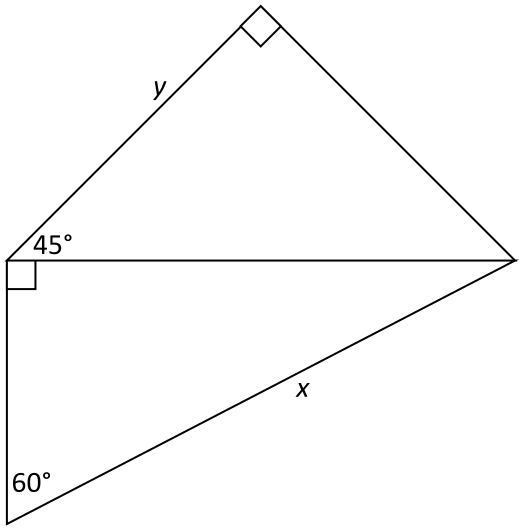
Question 4 **[3 marks]**

In the diagram, *O* is the centre of the circle with radius 6cm. Arc *JK* has a length measuring 2π cm. Determine the exact area of sector *OJK.*



Question 5 **[4 marks]**

Using the diagram below show that $y = \dfrac{\sqrt{3}\,x}{2\sqrt{2}}$



End of Section 1

CHISHOLM CATHOLIC COLLEGE

Year 11 Mathematics Methods

Test 1 2025

Section 2: Calculator - assumed

Name:	Time: 25 minutes minimum
Teacher:	Marks available: 27

Instructions:

- Answer all questions in the spaces provided.
- Up to 3 calculators are permitted.
- 2 double-sided A4 pages of notes are permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.
- Diagrams are not drawn to scale.

Question 5 **[1, 2, 2, 2 = 7 marks]**

A triangle WXY has $w = 45$ cm, $y = 34$ cm and an area of 739 cm^2.

(a) Sketch a triangle to show this information.

(b) Determine the size of $\angle X$.

(c) Show that $x \approx 63$ cm.

(d) Show that $\angle Y \approx 31°$.

Question 6 **[2 marks]**

The sides adjacent to the right-angle in a right triangled triangle have lengths 65 cm and72 cm.

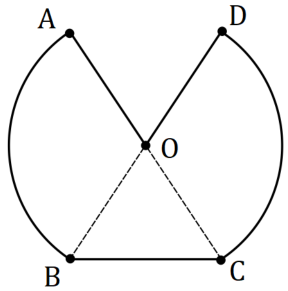
If the smallest angle in the triangle is α, determine an exact value for

(i) $\tan\alpha$.

(ii) $\sin(90° - \alpha)$.

Question 7 **[4, 3 = 7 marks]**

In shape $OABCD$ below, $\angle AOB = \frac{13\pi}{20}$ and AC, BD are diameters of the circle with centre O and radius 42 cm.

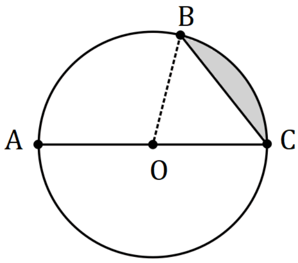


(a) Calculate the perimeter of $OABCD$.

(b) Calculate the area of $OABCD$.

Question 8 **[4, 4 = 8 marks]**

(a) The circle shown has centre O and diameter AC of length 50 cm. Determine the shaded area given that $2 \times \angle AOB = 3 \times \angle BOC$.



i) Show that $\angle BOC = \frac{2\pi}{5}$

ii) Determine the shaded area

(b) A sector of a circle has a perimeter of 112 cm and an area of 735 cm^2. Determine the radius of the circle.

Question 9 **[3 marks]**

Calculate, to the nearest degree, the acute angle between the line $y = 1.5x - 4$ and the line $y = -0.5x + 4$.